



School of Computer Science

COMP47470: Big Data Programming

**Estimations in Scalable Data System
Design**

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In this tutorial, you will solve exercises that involve estimating a scalable data system features (*disk storage*, *main memory size*, and *performance*) using simple estimation techniques.

The student is recommended to review **Lecture 2 on Introduction to Big Data**. Specifically, the topics on units for data volumes, times, availability and latency numbers every programmer should know.

Latency numbers every programmer should know

https://colin-scott.github.io/personal_website/research/interactive_latency.html

Note some of the exercises are simplified versions of the problems described in the resources below:

The System Design Primer

<https://github.com/donnemartin/system-design-primer>

System Design Interview – An insider's guide

Volumes 1 & 2

By Alex Hu

We start with an example problem (EP).

EP). We want to design a *search autocomplete system*. In this system, as you type the words, one or more results are presented to you. A system returns **top k** (for example, **top 5**) suggestions based on historical query frequency (popularity). We will study *sketching* and *approximation* algorithms to solve the **top k** problem in lectures on streams (Week 11).

Supposing we have gathered the following requirements:

1. There will be **10** million daily active users.
2. A user performs **10** searches on average per day.
3. Each search has **25** bytes.
4. The browser sends **25** requests to the backend for suggestions on average for each and every search query.
5. **20%** of the queries per day are new.

EPQ1). What is the amount of new data added to storage per day?

A). $10,000,000 * 10 * 25 * 20\% = 500,000,000 \text{ B} \approx 0.48 \text{ GB}$

EPQ2). What is the number of requests per second (QPS) to the backend?

A). $10,000,000 * 10 * 25 / 24 / 3600 \approx 29,000$ QPS

P1). We want to design a cloud storage service (something similar to MS OneDrive, Google Drive, and DropBox). Cloud storage service is a file storage and synchronization service that allows you to store and share files (videos, images, docs, spreadsheets) in the cloud.

Supposing we have the following requirements:

1. The service will have 50 million signed up users and 10 million daily users.
2. Users get 10 GB of storage.
3. Users upload on average 2 files per day. The average file size is 500 KB.

Q11). What is the total amount of storage needed (in PB)?

Q12). What is the number of files uploaded per second (QPS)?

Q13). What is the total amount of new storage needed (in PB)?

P2). We want to design a scalable database to support a new mobile application called *Nearby Friends*, which displays a list of friends who are geographically close by.

Supposing we have the following requirements:

1. 100 million users (daily active users) use the app every day.
2. The number of users simultaneously using the app is 10% of the total daily active users.
3. On average, a user has 400 friends who use the app. So, roughly 10% of these friends are online and nearby.
4. Users report their locations every 30 seconds.

Q21). What is the QPS (location updates per second) sent to the backend?

Q22). What is the QPS (location updates per second) forwarded by the backend?

P3). We want to design a scalable email service (similar to Gmail and Yahoo Mail). Supposing we have the following requirements:

1. The service must support 1 billion users.
2. The average number of emails a person sends per day is 10.
3. The average number of emails a person receives per day is 40.
4. The average size of metadata (excluding attachments) is 50 KB.
5. 20% of emails contain an attachment and the average attachment is 500 KB.

Q31). What is the QPS (number of emails per second) for sending emails?

Q32). What is the storage required for metadata in 1 year?

Q33). What is the storage required for attachments in 1 year?

P4). We want to design a scalable ad click event aggregation system (similar to Google). Following are the requirements:

1. The system must support 1 billion daily active users.
2. On average each user clicks 1 ad per day.
3. A single ad click event occupies 0.1 KB storage.

Q41). What is the monthly storage requirement of ad clicks (assume 30 days in a month)?

P5). We want to design a scalable web crawler, which is used for search engine indexing, web mining and web monitoring (similar to Googlebot).

Following are the requirements:

1. A web crawler downloads 1 billion web pages every month.
2. The average web page size is 500 KB.

Q51). What is the storage needed to store 5-year content?